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LAB SET-5

9. Write a Pandas program to create a Pivot table and find the total sale amount region wise, manager wise, sales man wise.

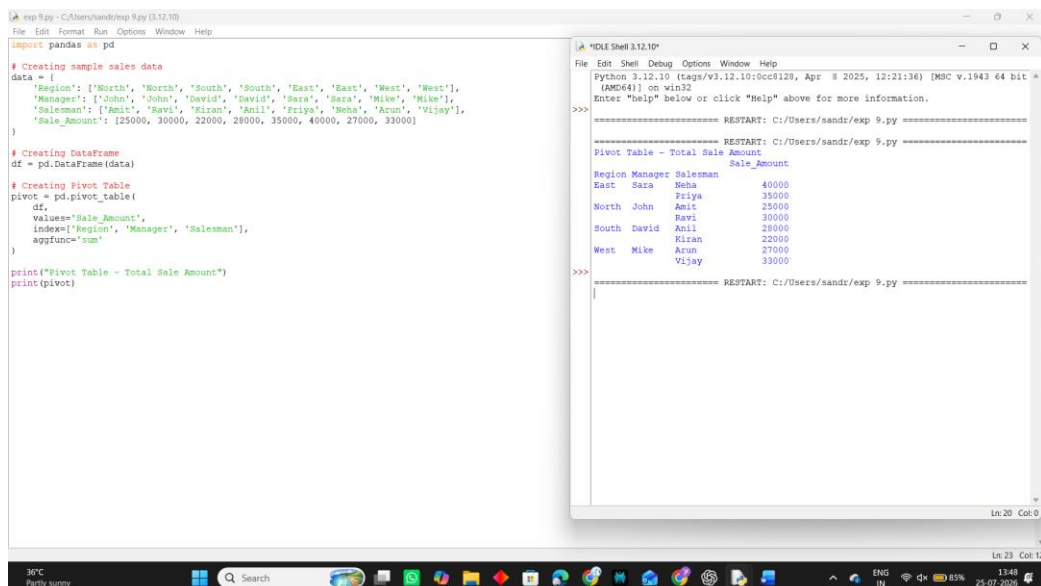
Aim

To create a Pivot Table using Pandas and find the total sale amount based on Region, Manager, and Salesman.

Procedure

1. Import the Pandas library.
2. Create a sales dataset.
3. Convert the dataset into a DataFrame.
4. Use the `pivot_table()` function.
5. Group the data by Region, Manager, and Salesman.
6. Calculate the sum of the `Sale_Amount`.
7. Display the Pivot Table.

OUTPUT:



The screenshot shows a Python IDE with two windows. The left window displays the code for creating a Pivot Table, and the right window shows the output of the code.

```
# Creating sample sales data
data = {
    'Region': ['North', 'North', 'South', 'South', 'East', 'East', 'West', 'West'],
    'Manager': ['John', 'John', 'David', 'David', 'Sara', 'Sara', 'Mike', 'Mike'],
    'Salesman': ['Amit', 'Ravi', 'Kiran', 'Anil', 'Priya', 'Meha', 'Arun', 'Vijay'],
    'Sale_Amount': [25000, 30000, 22000, 28000, 35000, 40000, 27000, 33000]
}

# Creating DataFrame
df = pd.DataFrame(data)

# Creating Pivot Table
pivot = pd.pivot_table(
    df,
    values='Sale_Amount',
    index='Region', 'Manager', 'Salesman'],
    aggfunc='sum'
)

print("Pivot Table - Total Sale Amount")
print(pivot)
```

The output window shows the following Pivot Table:

Pivot Table - Total Sale Amount			Sale_Amount
Region	Manager	Salesman	
East	Sara	Meha	40000
		Priya	35000
North	John	Amit	25000
		Ravi	30000
South	David	Anil	28000
		Kiran	22000
West	Mike	Arun	27000
		Vijay	33000

Result

Thus, the Pandas program was executed successfully to create a Pivot Table and find the total sale amount region-wise, manager-wise, and salesman-wise.

10. Create a dataframe of ten rows, four columns with random values. Write a Pandas program to highlight the negative numbers red and positive numbers black.

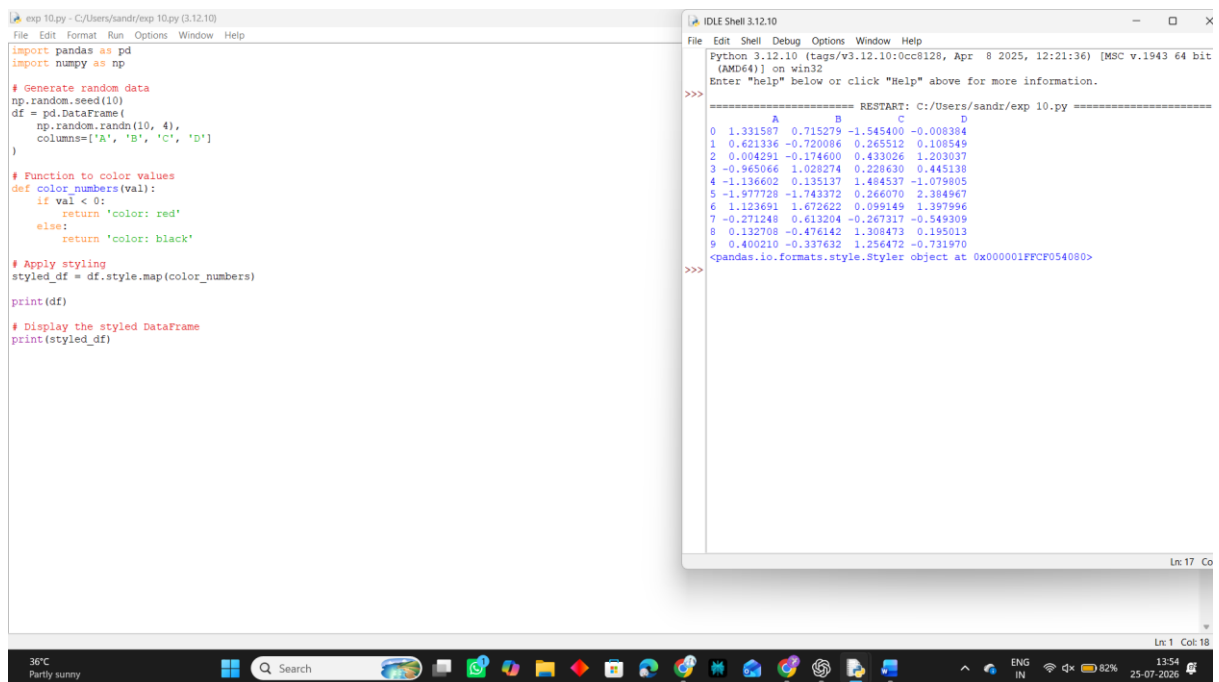
Aim

To create a Pandas DataFrame with random values and highlight negative numbers in red and positive numbers in black using DataFrame styling.

Procedure

1. Import the Pandas and NumPy libraries.
2. Generate random values using NumPy.
3. Create a DataFrame with 10 rows and 4 columns.
4. Define a function to apply colors based on the value.
5. Use the `style.applymap()` method to format the DataFrame.
6. Display the styled DataFrame.

Output:



The screenshot shows a Python IDE with two windows. The left window displays the source code for a Pandas program that generates a DataFrame with 10 rows and 4 columns of random values. It defines a function `color_numbers` to map values to colors (red for negative, black for positive) and applies this styling to the DataFrame. The right window shows the execution output, which is a table of 10 rows and 4 columns. The values are formatted with their corresponding colors: negative numbers are in red and positive numbers are in black.

```
import pandas as pd
import numpy as np

# Generate random data
np.random.seed(10)
df = pd.DataFrame(
    np.random.randn(10, 4),
    columns=['A', 'B', 'C', 'D']
)

# Function to color values
def color_numbers(val):
    if val < 0:
        return 'color: red'
    else:
        return 'color: black'

# Apply styling
styled_df = df.style.map(color_numbers)
print(df)

# Display the styled DataFrame
print(styled_df)
```

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/sandr/exp 10.py =====
   A      B      C      D
0  1.331587  0.715279 -1.545400 -0.008384
1  0.621336 -0.720086  0.265512  0.108549
2  0.004291 -0.174600  0.433026  1.203037
3 -0.965066  1.028274  0.228630  0.445138
4 -1.136602  0.135137  1.484537 -1.079805
5 -1.977728 -1.743372  0.266070  2.384967
6  1.123681  1.672622  0.099149  1.397996
7 -0.271248  0.613204 -0.267317 -0.549309
8  0.132708 -0.476142  1.308473  0.195013
9  0.400210 -0.337632  1.256472 -0.731970
<pandas.io.formats.style.Styler object at 0x000001FFCF054080>
>>>
```

Result

Thus, the Pandas program was executed successfully to create a DataFrame with random values and highlight negative numbers in red and positive numbers in black.